

MPC-MAP Assignment No. 2 - Report

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Task 1

Two maps have been created for testing motion control: one simpler and one more complex, each 15x15 in size.

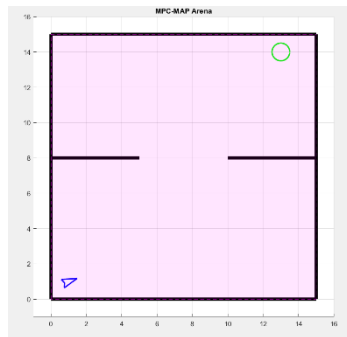


Figure 1 – week3 map 1

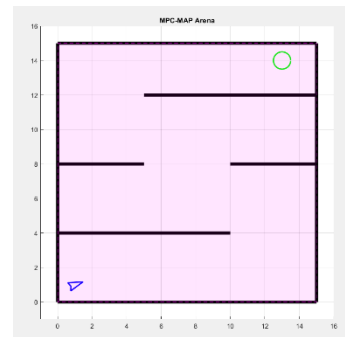


Figure 1 – week3 map complex

Task 2

Three defined paths on the created map. The first path is a straight line, the second is a straight line with added sine function, and the third consists of two arcs

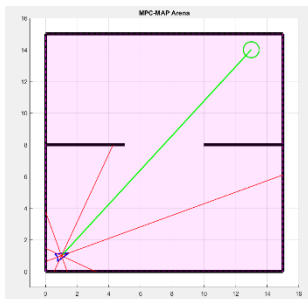


Figure 3 – Path 1 – straight line

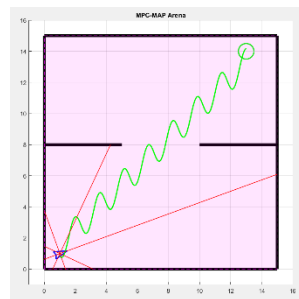


Figure 4 – Path 2 – with added sine

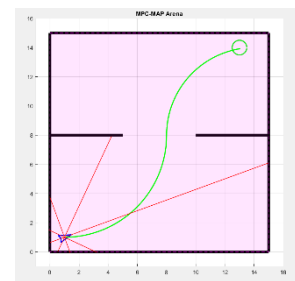


Figure 5 – Path 3 - arcs

Task 3

To make the robot follow the path, a feedback linearization path-following algorithm has been implemented and tested on the paths created in Task 2 and on the path on the map in Figure 2. The feedback linearization algorithm has two parameters, kappa and epsilon. Kappa is a gain of distance error of point P from the path and acts as a P controller. The higher the kappa, the higher the control action in response to the error. Epsilon represents the look-ahead distance (distance between robot and point P). A larger epsilon gives a smoother but less accurate path following. The parameters were set to kappa = 6 and epsilon = 0.25.

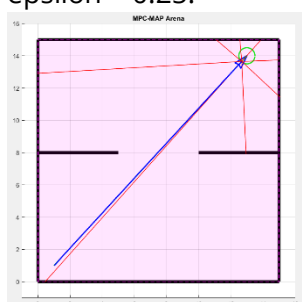


Figure 6 – Line path following

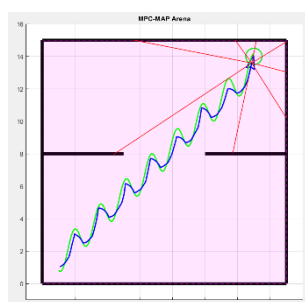


Figure 7 – Sine path following

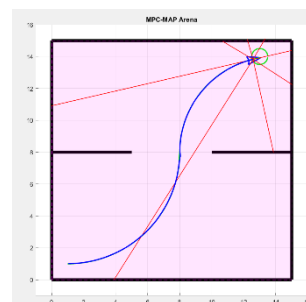


Figure 8 – Arc path following

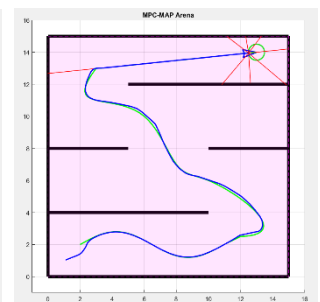


Figure 9 – Combined path following